EQUIDIMENSIONALITY OF A RELATIVE CYCLE IS A ZARISKI-LOCAL CHECK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition of equidimensional relative cycles (more precisely: the condition that alpha is equidimensional means the closure of its support has relative dimension at most r over S; in-source context, Definition definition-equidimensional at lines 1537-1547 of the same chapter, no reference in the counted span). restated at proof lines 1594-1596. The definition being unwound; the proof reduces the local check to this condition.
- The standard fact that taking closures commutes with restriction to opens (stated unnamed at proof line 1596). stated at proof line 1596. The single ingredient: closure of the support restricted to an open equals the closure of the restricted support, which makes the support-closure condition local on X.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Assume S locally Noetherian and f locally of finite type. Let $r \geq 0$ be an integer. Let α be a relative r-cycle on X/S. Then to check that α is equidimensional we may work Zariski locally on X and S.*

Proof. Namely, the condition that α is equidimensional just means that the closure of the support of α has relative dimension $\leq r$ over S. Since taking closures commutes with restriction to opens, the lemma follows (small detail omitted). □

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